

Lia Schmid

B.Sc. COGNITIVE SCIENCE · M.Sc. SCIENTIFIC COMPUTING

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Summary

With a Master's in Scientific Computing and hands-on experience building and coordinating data pipelines for AI/ML systems, including EEG-based decoding models. I care about getting the fundamentals right by setting and respecting data-quality standards, keeping cross-institutional and cross-border collaborations running smoothly across academic, clinical, and technical teams. I adapt quickly to new environments and integrate easily into different teams and ways of thinking. I'm fluent in German, English, Italian, and Portuguese, and I'm drawn to work that sits at the intersection of data infrastructure, ML systems and people depending on them.

Core Competences

- Data Operations & Quality Assurance: defining data collection protocols, quality benchmarks, and documentation for ML/AI training pipelines
- Cross-Functional & Cross-Institutional Coordination: bridging AI/ML engineers, clinical experts, and academic partners across countries
- Technical Toolkit: Python, R, MATLAB, SQL (MySQL), Git, Singularity/containerization, ML pipeline development
- Training Design & Onboarding: building structured curricula and onboarding technical cohorts (M.Sc. students, course participants)
- Languages: German (fluent/native), Portuguese (fluent/native), English (fluent), Italian (fluent)

Projects

Development of a EEG BCI Solution: EEG Data Pipeline & Model Development

Milan, Italy

BRAINCAPNET

04/2024 - today

- Build and led the EEG data pipeline: acquisition to model-ready datasets: BCI collaboration with dotdotdot and Fondazione TOG: giving locked-in and neurodegenerative children a channel to communicate
- Built and trained subject-independent EEG-decoding models translating brain signals into user-intent; results accepted at ICMI 2026

Cross-Institutional Data Governance: (Heidelberg / Milan / Policlinico di Milano)

Milan, Italy

MASTER'S THESIS PROJECT: AUTOMATIC SEGMENTATION OF CHOROID PLEXUS IN NEURODEGENERATIVE DISEASES

01/2024 - 11/2024

- Coordinated data-sharing arrangements and quality requirements for a clinical imaging dataset across three institutions and two countries, aligning computer scientists, neurologists, and radiologists on a shared ground-truth standard.
- Delivered a novel ground-truth methodology for automatic segmentation, later published across peer-reviewed venue.

Work Experience

Perceptual computing and Human Sensing Lab (PHuSe Lab), University of Milan

Milan, Italy

RESEARCH FELLOW (POSTGRADUATE RESEARCHER) IN BRAIN-COMPUTER INTERFACES (EEG)

04/2025 - today

- Conduct research on EEG-based Brain-Computer Interfaces (BCIs).
- Develop and evaluate machine learning models for decoding brain signals and user intent.
- Coordinate scope and milestones with faculty and cross-lab collaborators, translating research goals into a structured data and model roadmap.

Catholic University of the Sacred Heart

Milan, Italy

TUTOR FOR COMPUTER PROGRAMMING AND DATABASE SYSTEMS

02/2026 - 04/2026

- Designed and delivered a structured Python and DBMS (MySQL, DBeaver) exercitation for M.Sc. Data Science and AI for Business students

Engineering Mathematics and Computing Lab (EMCL), University of Heidelberg

Heidelberg, Germany

RESEARCH ASSISTANT

05/2023 - 02/2024

- Co-organizing a project and building technical infrastructure (Singularity container) to analyze morphological structures

Eberhard Karls University

Tübingen, Germany

TUTOR IN MATHEMATICS FOR COMPUTER SCIENCE

10/2020 - 12/2020

- Preparation course for freshmen.
- Assisting new students in achieving a consistent mathematical foundation while building social connections.

Vision and Cognition Lab, Werner Reichardt Centre for Integrative Neuroscience

Tübingen, Germany

RESEARCH INTERNSHIP/ RESEARCH ASSISTANT

03/2020 - 12/2020

- Safety briefing on an fMRI device.
- Comparing hierarchical processing of the visual cortex to layers of convolutional neural networks (python, MATLAB).

Eberhard Karls University

TUTOR IN MATHEMATICS FOR COMPUTER SCIENCE

- Covering topics of calculus, linear algebra, optimization, and discrete mathematics.

Tübingen, Germany

10/2020 - 12/2020

Eberhard Karls University

STUDENT COUNSELING FOR COGNITIVE SCIENCE

- Providing peer-to-peer guidance on course selection, schedule planning, and exam requirements.
- Assisting students with organizational questions, including housing and funding options.

Tübingen, Germany

10/2018 - 03/2021

Education

Master of Science in Scientific Computing

RUPRECHT-KARLS-UNIVERSITÄT

- Thesis: "Automatic Segmentation of Choroid Plexus in Neurodegenerative Diseases: Leveraging the T1xFLAIR Product as a Novel Ground Truth Modality" (Collaboration: University of Milan, Policlinico of Milan and University of Heidelberg).
- Study abroad (4EU+ Erasmus programme): Year abroad at Università degli Studi di Milano, Milan, Italy (09/2022 – 09/2023)
- Final Degree: 1,1

Heidelberg, Germany

04/2021 – 11/2024

Bachelor of Science in Cognitive Science

EBERHARD KARLS UNIVERSITÄT

- Study abroad (Erasmus programme): Semester abroad at Universidade Nova de Lisboa, Lisbon, Portugal (09/2019 – 01/2020)
- Final Degree: 1,47

Tübingen, Germany

10/2017 – 04/2021

(European) Sport Nutritionist

ACCADEMIA FITNESS ITALIANA

- Certified Sports Nutritionist: Trained in human physiology, sports psychology, and the application of professional nutritional plans.
- Client Management Specialist: Developed practical skills for designing dietary strategies and managing professional athlete-client relations.

Milan, Italy

12/2025 – 06/2026

Publications

- Ghezzi, O., Schmid, L., D'Amelio, A., Boccignone, G., Lanzarotti, R.: "Where the Eyes Move, the Brain Flows: Brain Decoding During Active Reading from Travelling-Wave Geometry" (submitted: TCML 2026).
- Schmid, L., Burger, J., D'Amelio, Lanzarotti, R.: "Investigating Foundation Models, Disentanglement and Latent Alignment for Subject-Independent EEG Learning" (ICMI 2026).
- Schmid, L., Facchi, G., Agnelli, F., Bocca, G., Sacchi, L., Lanzarotti, R.: "Choroid Plexus Segmentation in MRI Using the Novel T1xFLAIR Modality and PSU-Mamba: Projective Scan U-Mamba Approach" (PRL Elsevier 2025).
- Sacchi, L., Arcaro M., Bocca, G., Schmid, L., Carandini T., Ghezzi L., Pintus M., Pietroboni A. M., Fenoglio C., Serpente M., Conte G., Triulzi F., Lanzarotti R., Dolci C., Galimberti, D., and Arighi A. "Klotho levels in the cerebrospinal fluid are associated with choroid plexus enlargement in neurodegeneration: a preliminary study" (Frontiers in Aging Neuroscience)
- Agnelli, F., Ghezzi, O., Blandano, G., Burger, J., Facchi, G., and Schmid, L. Enhancing 3D Face Analysis Using Graph Convolutional Networks with Kernel-Attentive Filters. Manuscript submitted to The 40th ACM/SIGAPP Symposium on Applied Computing (SAC 2025).
- Agnelli, F., Blandano, G., Burger, J., D'Amelio, A., Facchi, G., Ghezzi, O., ... & Schmid, L. (2025, September). EEG-Based Mental Stress Detection: A Comparative and Explainable Study Across Tasks and Subjects (ICIAP 2025).

Extracurricular Activity

Wizz Air Marathon Milano 2026

MEDAL ENGRAVING AND PACERS BAGGAGE

- Medal Engraving, pacers baggage, dog run support

Milan, Italy

04/2026

Milano Cortina 26

EVENT SERVICE VOLUNTEER OLYMPIC GAMES MILANO CORTINA 26

- Event Service at Olympic Village, Milan

Milan, Italy

01/2026 - 02/2026

University of Tübingen

MEMBER OF THE STUDY COMMISSION OF COGNITIVE SCIENCE

- Development and evaluation of study plan, Voting of students matters

Tübingen, Germany

10/2018 - 04/2021

FSK Tübingen

MEMBER OF THE STUDENT COUNCIL OF COGNITIVE SCIENCE

- Representing student interests, Communication between students and instructors, Organizing social events

Tübingen, Germany

10/2018 - 04/2021